Calculus I

Tutorial #9 (25–29 Nov, 2024)

Topics:

- Basic integral formulas
- Integration by parts
- Trigonometric integrals

Problem 1. Evaluate each integral using any algebraic method or trigonometric identity you think is appropriate, and then use a substitution to reduce it to a standard form.

a)
$$\int_0^1 \frac{16x}{8x^2+2}\,dx.$$

b)
$$\int \frac{e^{-\cot z}}{\sin^2 z}\,dz.$$

c)
$$\int \frac{d\theta}{\sqrt{2\theta-\theta^2}}.$$

d)
$$\int \frac{2\,dx}{x\sqrt{1-4\ln^2 x}}.$$

e)
$$\int e^{z+e^z}\,dz.$$

Problem 2. *(Using different substitutions).* Show that the integral
$$\int ((x^2-1)(x+1))^{-2/3}\,dx$$
can be evaluated with any of the following substitutions.

(a) $u = 1/(x+1)$.

(b) $u = ((x-1)/(x+1))^k$ for $k = 1,\ 1/2,\ 1/3,\ -1/3,\ -2/3$, and -1.

(c) $u = \tan^{-1} x$.

(d) $u = \tan^{-1} \sqrt{x}$.

(e) $u = \tan^{-1}((x-1)/2)$.

(f) $u = \cos^{-1} x$.

(g) $u = \cosh^{-1} x$.

Problem 3. Evaluate the integrals using integration by parts.

(a)
$$\int (r^2 + r + 1)e^r \, dr.$$

(b)
$$\int e^\theta \sin\theta \, d\theta.$$

Problem 4. Evaluate the integrals. Some integrals do not require integration by parts.

(a)
$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} \, dx.$$

(b)
$$\int_0^{\pi/2} x^3 \cos(2x) \, dx.$$

Problem 5. Use integration by parts to establish the reduction formula.

(a)
$$\int x^n \cos(x) \, dx = x^n \sin(x) - n \int x^{n-1} \sin(x) \, dx.$$

(b)
$$\int x^m (\ln x)^n \, dx = \frac{x^{m+1}}{m+1}(\ln x)^n - \frac{n}{m+1} \int x^m (\ln x)^{n-1} \, dx, \quad m \neq -1.$$

Problem 6. *(Powers of Sines and Cosines)* Evaluate the integral
$$\int 8\cos^3 2\theta \, \sin 2\theta \, d\theta.$$

Problem 7. *(Integrating Square Roots)* Evaluate the integral
$$\int_0^\pi \sqrt{1 - \cos^2\theta} \, d\theta.$$

Problem 8. *(Powers of Tangents and Secants)* Evaluate the integral

$$\int \sec^2 x \, \tan^2 x \, dx.$$

Problem 9. *(Products of Sines and Cosines)* Evaluate the integral

$$\int \cos 3x \, \cos 4x \, dx.$$

Problem 10. *(Assorted Integrations).* Use any method to evaluate the integrals.

(a)

$$\int \frac{\sec^3 x}{\tan x} \, dx.$$

(b)

$$\int x \, \sin^2 x \, dx.$$

THE END

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Answer to selected problems

1. (a) $\ln 5$ (b) $e^{-u} + C$ (c) $\sin^{-1}(\theta - 1) + C$ (d) $\sin^{-1}(2 \ln x) + C$ (e) $e^{e^z} + C$
3. (a) $e^r(r^2 - r + 2) + C$ (b) $\frac{1}{2}e^\theta(\sin\theta - \cos\theta) + C$
10. (a) $\sec x - \ln|\csc x + \cot x| + C$ (b) $\dfrac{x^2}{4} - \dfrac{x \sin 2x}{4} + \dfrac{\cos 2x}{8} + C$